A study of experimental privacy budget quantification in quantum differential privacy simulations

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ABSTRACT 150-250 palabras

INDEX TERMS Differential privacy, Quantum computing, Quantum noise

I. INTRODUCTION

ESTA va a ser la introduccion a la tesis. En esta sección pienso hablar de la motivación, la importancia de la privacidad diferencial, basicamente muy similar al documento de propuesta

II. BACKGROUND

On this section we will introduce all the background information that the reader might find usefull for understanding the investigation. This secction wwill introduce concepts, equations and any other piece of information to this end.

A. QUANTUM INFORMATION AND COMPUTATION

Definition 2.1: (Quantum states [1]). A quantum state of the system X is described by a density matrix ρ whose entries are complex numbers and whose indices (for both its rows and columns) have been placed in correspondence with the classical state set Σ.

Definition 2.2: (Mixed state [2]). A mixed state of a quantum system is an ensemble

$$\mathbb{E} = \{(p_1, |\psi_1\rangle), ..., (p_k, |\psi_k\rangle)\}$$

which in turn can be represented by the density operator

$$\rho_{\mathbb{E}} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$$

thus the resulting $\rho_{\mathbb{E}}$ will be the density matrix representation of the ensemble.

Definition 2.3: (Quantum channel [1]). A quantum channel describes a linear mapping over density matrices.

Quantum channels are important as they are the generalization of an operation over a quantum state, and in the case of this project we need to use operations such as the depolarizing channel, which is defined thanks to this general definition.

Definition 2.4: (Bloch sphere [1]). The Bloch sphere is a useful geometrical representation for a single qubit.

Definition 2.5: (Trace distance [2]). It is the distance between two density operators and it is defined by:

$$\tau(\rho, \sigma) = \frac{1}{2} Tr|\rho - \sigma|$$

where $Tr(x)$ is the trace of the density matrix x.

Another useful form to define the trace distance in terms of vectors, by Nielsen & Chuang [3], is

Definition 2.6: (Trace distance [3]). It is the distance between two qubit vectors u and v defined by:

$$\tau(u, v) = \frac{1}{2}\|u - v\|$$

Definition 2.7: (Quantum measurement [2]). It is the way to extract information from a quantum system. Mathematically, a measurement is modeled as a set of matrices $M = \{K_m\}$ with the condition of

$$\sum_m K_m^\dagger K_m = I$$

with I being the identity of the density operator (i.e., for a qubit, I is the 2×2 identity matrix).

When talking about quantum differential privacy below, it will not really matter what the quantum state collapses into. Given this liberty, we can define a POVM (Positive Operator-Valued Measure) which is a simplification over the measurement where we get a set of matrices $M = \{M_m\}$ with $M_m = K_m^\dagger K_m \ \forall m$. This is possible since we only care about the probabilities, and so $p_m = Tr(K_m \rho K_m^\dagger) = Tr(M_m \rho)$.

Definition 2.8: (Quantum noise [2]). It is an inherent physical process that occurs on the quantum state that causes it to change in some way.

For the purposes of this investigation we are interested in three types of noise: generalized amplitude damping, phase-amplitude damping and depolarizing noise. Each one of these types represents a unique physical process which is summarized in greater detail in [4] and [2]. The definitions of the noise channels are:

Definition 2.9: (Generalized amplitude damping mechanism [2]). The generalized amplitude damping channel operation ($\mathcal{E}_{GAD}$) over a density matrix ρ is defined as

$$\mathcal{E}_{GAD}(\rho) = \sum_{k=0}^{3} E_k \rho E_k^\dagger$$

in the 2-dimensional Hilbert space

where

$$E_0 = \sqrt{\eta}\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \qquad E_1 = \sqrt{\eta}\begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix},$$
$$E_2 = \sqrt{1-\eta}\begin{pmatrix} \sqrt{1-\gamma} & 0 \\ 0 & 1 \end{pmatrix}, \quad E_3 = \sqrt{1-\eta}\begin{pmatrix} 0 & 0 \\ \sqrt{\gamma} & 0 \end{pmatrix}$$

and η and γ are two parameters

and the generalized amplitude damping mechanism ($\mathcal{M}_{GAD}$) is the mapping

$$\rho \mapsto \mathcal{M}_{GAD} = \mathcal{E}_{GAD}(\rho)$$

Definition 2.10: (Phase damping channel [2]). The phase damping channel operation ($\mathcal{E}_{PD}$) over a density matrix ρ is defined as

$$\mathcal{E}_{PD}(\rho) = \sum_{k=0}^{1} E_k \rho E_k^\dagger$$

in the 2-dimensional Hilbert space

where

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}$$

with λ being a parameter

Definition 2.11: (Phase-amplitude damping mechanism [2]). The phase-amplitude damping channel operation ($\mathcal{E}_{PAD}$) over a density matrix ρ in the Hilbert space $\mathcal{H}_2$ is defined as the mapping

$$\rho \mapsto \mathcal{M}_{PAD} = \mathcal{E}_{GAD}(\mathcal{E}_{PD}(\rho))$$

Definition 2.12: (Depolarizing channel mechanism [2]). The depolarizing channel operation ($\mathcal{E}_{DEP}$) over a density matrix ρ with a Hilbert space dimension D and I the $D \times D$ identity matrix can be represented by the quantum operation

$$\mathcal{E}_{DEP}(\rho) = \frac{pI}{D} + (1-p)\rho$$

where p is the probability parameter

and the depolarizing mechanism ($\mathcal{M}_{DEP}$) is the mapping

$$\rho \mapsto \mathcal{M}_{DEP} = \mathcal{E}_{DEP}(\rho)$$

B. DIFFERENTIAL PRIVACY

Differential privacy describes a promise that is made by a data holder or curator to a data subject: "You will not be affected, adversely or otherwise, by allowing your data to be used in any study or analysis, no matter what other studies, data sets, or information sources, are available" [5]. In other words, differential privacy is the process that either a data curator or a data holder must do in order to ensure that a single individual's data cannot be extracted from the study as a whole. In *The Algorithmic Foundations of Differential Privacy*, Dwork & Roth formalize differential privacy as

Definition 2.13: (Differential privacy [5, Def. 2.4]). A randomized algorithm $\mathcal{M}$ with domain $\mathbb{N}^{|\mathcal{X}|}$ is (ϵ, δ)-differentially private if for all $S \subseteq \mathrm{Range}(\mathcal{M})$ and for all $x, y \in \mathbb{N}^{|\mathcal{X}|}$ such that $\|x - y\|_1 \leq 1$:

$$\Pr[\mathcal{M}(x) \in S] \leq \exp(\epsilon) \Pr[\mathcal{M}(y) \in S] + \delta,$$

where the probability space is over the coin flips of the mechanism $\mathcal{M}$. If $\delta = 0$, we say that $\mathcal{M}$ is ϵ-differentially private.

Here a database $x \in \mathbb{N}^{|\mathcal{X}|}$ is represented by its histogram over the universe $\mathcal{X}$ of possible rows (x_i is the number of rows of type i), and $\|x - y\|_1 = \sum_i |x_i - y_i|$ is the number of records in which x and y differ [5, Def. 2.3].

In our investigation we are interested in the privacy guarantee ϵ when $\delta = 0$. The value of ϵ can be interpreted as a privacy guarantee from the differentially private algorithm as it implies that adding or removing one record (i.e., adding or removing a participant) in the dataset x will change the probability of any output by a factor of at most e^ϵ. In other words, an adversary who wants to analyze the dataset will not be able to reliably tell if a person was included in the dataset or not.

In order to bound this ϵ, Dwork & Roth define the privacy loss of an output as:

Definition 2.14: (Privacy loss [5, Sec. 2.3]). Let $x, y \in \mathbb{N}^{|\mathcal{X}|}$ be adjacent databases ($\|x - y\|_1 \leq 1$) and let ξ be an output of $\mathcal{M}$. The privacy loss incurred by observing ξ is

$$\mathcal{L}^{(\xi)}_{\mathcal{M}(x)\|\mathcal{M}(y)} = \ln\left(\frac{\Pr[\mathcal{M}(x) = \xi]}{\Pr[\mathcal{M}(y) = \xi]}\right)$$

This privacy loss will be used throughout this investigation to experimentally measure ϵ_{exp} in the experiments explained in Section V.

For the purpose of our investigation we were interested in the two settings that can be thought of when working on differential privacy: **Central** and **Local**. The central setting (or curator model, as explained in [5] and by Definition 2.13) explains how a trusted third party receives the unprotected data, applies differentially private methods to the data and

releases the outputs computed from them. On the other hand, the local setting explains how each individual is responsible for applying a differentially private method to their own data and then releases the result. In other words, it is a special case of Definition 2.13 where the database is of size 1 [5, Def. 12.1].

C. QUANTUM DIFFERENTIAL PRIVACY

When it comes to quantum differential privacy, the promise that this mechasims are trying to offer to the data holder will not change. However the means that will be taken to reach that promise will. In [2] They define the concept of quantum differential privacy as follows

Definition 2.15: (Quantum differential privacy [2]). Let $d \in (0, 1]$ and $\epsilon, \delta > 0$ be three constants. A quantum operation $\mathcal{E}$ is (ϵ, δ)-differentially private if for every POVM $M = \{M_m\}$, for all outputs $S \subseteq Out(M)$ and for all inputs ρ, σ such that $\tau(\rho, \sigma) \leq d$, it holds that:

$$Pr[\mathcal{E}(\rho) \in_M S] \leq e^{\epsilon} Pr[\mathcal{E}(\sigma) \in_M S] + \delta$$

There are many ways to obtain ϵ in theory; however, Zhou and Ying proposed the following algorithm to obtain a quantity called proportional distance which can help obtain ϵ.

Definition 2.16: (Proportional Distance [2]). For two density operators ρ and σ, their proportional distance is defined as:

$$PD(\rho, \sigma) = \sup_{\{M_m\}, m} \max \left(\ln \frac{q_m}{p_m}, \ln \frac{p_m}{q_m} \right)$$

where the supremum is over all POVMs $\{M_m\}$ and all possible outcomes m,

$$p_m \equiv Tr[\rho M_m], \quad q_m \equiv Tr[\sigma M_m]$$

are the probabilities of obtaining outcome m when the measurement is performed on ρ, σ, respectively, and we make the conventions $\frac{0}{0} = 1$, $\frac{1}{0} = +\infty$, $\ln 0 = -\infty$, $\ln +\infty = +\infty$.

Proposition 2.1 (Zhou and Ying [2]):

$$\mathrm{PD}(\rho, \sigma) = \sup_{|\psi\rangle} \max \left(\ln \frac{\langle\psi| \rho |\psi\rangle}{\langle\psi| \sigma |\psi\rangle}, \ln \frac{\langle\psi| \sigma |\psi\rangle}{\langle\psi| \rho |\psi\rangle} \right)$$

where the supremum is over all pure states.

Definition 2.17: (dPD [2]). For two density operators ρ and σ,

$$\mathrm{dPD}(\rho, \sigma) = \sup_{0 < M \leq I} \ln \frac{\mathrm{Tr}(M\rho)}{\mathrm{Tr}(M\sigma)}$$

so that

$$\mathrm{PD}(\rho, \sigma) = \max \left(\mathrm{dPD}(\rho, \sigma), \mathrm{dPD}(\sigma, \rho) \right). \tag{1}$$

As the reader can observe, dPD is nothing more than the privacy loss that [5] defined, and thus if we find dPD then we can find ϵ. Algorithm 1 [2] provides a linear programming algorithm in order to compute dPD between two density operators σ and ρ with an accuracy of acc.

Algorithm 1 Computing dPD [2]

Require: density operators ρ, σ in a Hilbert space $\mathcal{H}$ of dimension D; desired accuracy acc

$r \leftarrow 0$, $r_{prev} \leftarrow \infty$
while $|r - r_{prev}| > acc$ **do**
  $r_{prev} \leftarrow r$
  $r \leftarrow \max_i(\rho_{ii}/\sigma_{ii})$
  $A \leftarrow \rho - r\,\sigma$
  Diagonalize $A = P\Lambda P^{\dagger}$ (P unitary, Λ diagonal)
  $\rho \leftarrow P^{\dagger}\rho P$
  $\sigma \leftarrow P^{\dagger}\sigma P$
end while
return $\ln r$

Theorem 2.1 (Zhou and Ying [2]): Let $\eta = 0.5$ in $\mathcal{E}_{GAD}$. For all inputs ρ and σ with $\tau(\rho, \sigma) \leq d$, the generalized amplitude damping mechanism $\mathcal{M}_{GAD}$ provides ϵ-differential privacy where

$$\epsilon = \ln \left[1 + \frac{2d\sqrt{1-\gamma}}{1-\sqrt{1-\gamma}} \right]$$

Theorem 2.2 (Zhou and Ying [2]): Let $\eta = 0.5$ in $\mathcal{E}_{GAD}$. If the parameter γ in $\mathcal{E}_{GAD}$ and λ in $\mathcal{E}_{PD}$ satisfy $\lambda \leq \gamma$, then for all inputs ρ and σ such that $\tau(\rho, \sigma) \leq d$, the phase-amplitude damping mechanism $\mathcal{M}_{PAD}$ provides ϵ-differential privacy where

$$\epsilon = \ln \left[1 + \frac{2d\sqrt{1-\gamma}\sqrt{1-\lambda}}{1-\sqrt{1-\gamma}\sqrt{1-\lambda}} \right]$$

Theorem 2.3 (Zhou and Ying [2]): For all inputs ρ and σ such that $\tau(\rho, \sigma) \leq d$, the depolarizing mechanism $\mathcal{M}_{DEP}$ in the D-dimension Hilbert space provides ϵ-differential privacy where

$$\epsilon = \ln \left[1 + \frac{1-p}{p}\, dD \right]$$

Corollary 2.1 (Zhou and Ying [2]): For each $i \in [k]$, let $\mathcal{M}_i : \mathcal{H}_i \to \mathcal{H}_i'$ be an ϵ_i-differentially private quantum algorithm, and assume that they are independent. Their combination

$$\mathcal{M}_{[k]} : \bigotimes_{i=1}^{k} \mathcal{H}_i \to \bigotimes_{i=1}^{k} \mathcal{H}_i'$$

is defined by

$$\mathcal{M}_{[k]}(\rho_1, \rho_2, \ldots, \rho_k) = \bigotimes_{i=1}^{k} \mathcal{M}_i(\rho_i)$$

for all density operators ρ_i in $\mathcal{H}_i$. Then $\mathcal{M}_{[k]}$ is $\left(\sum_{i=1}^{k} \epsilon_i \right)$-differentially private.

III. RELATED WORK

It was important to figure out what the current investigative and academic situation was with regard to the fields of quantum information and computation and differential privacy. For this an investigation was made in which different databases were searched and analyzed to find out the number of papers that covered the topics mentioned before. The investigation found 61 relevant papers that focused on the topics and the specific distributions of each one are compiled in Figure 1.

As shown in Figure 1, the investigation yielded positive results as there were many academic papers that not only mentioned the topic but gave valuable research and insight on possible angles the project could leap into. Among the most interesting ones are: [6] in which a hybrid quantum machine learning framework was proposed to allow differential privacy in the model. [4] which tested different quantum differential privacy mechanisms along with proposing its own **Lifted** quantum differential privacy, which enhanced the randomness induced in the quantum states. [7] in which an extended analysis was made to characterize the protection quantum noise can offer in the setting of quantum-classical algorithms. Finally, in the study of Zhou and Ying [2], they define a framework of differential privacy in the quantum setting and examine how different noise mechanisms change the protection of a privacy budget ϵ.

IV. DATA PROCESSING AND ANALYSIS

This section details the data processing and analysis methodology that was used in order to prepare the classical Statlog (German Credit Data) dataset [8], [9] for use in the quantum setting. Firstly, we outline the dataset characteristics, then the variable selection procedures and, lastly, the quantum encoding procedures for the selected data.

A. DATASET DESCRIPTION

The selected dataset for this investigation is the Statlog (German Credit Data) dataset created by Professor Hans Hofmann [8]. This multivariate dataset aims to classify people as a good or bad credit risk. The original dataset is composed of 1000 instances that span 21 variables (7 numerical, 13 categorical, and the target).

In the original dataset created by Professor Hofmann, all non-numerical variables use an attribute code to define their characteristics (e.g., the status of a person whose checking account is less than 0 Deutsche Marks (DM) is A11). For that reason, it was decided that the investigation was going to use a curated version of the dataset [9] which transforms all the labels to their definition as a means of simplifying the readers' understanding of the dataset by eliminating that intermediate layer of label and category. Every mention of the dataset henceforth will refer to this curated version and not the original compiled by Professor Hofmann.

A simple data analysis concluded that the dataset did not have any problem in terms of completeness (as all rows include all 21 variables) or uniqueness (as there is no instance of an exact duplicate). For these reasons, no special cleaning procedure was done prior to variable selection.

B. VARIABLE SELECTION WITH PHIK

In order to hold quantum data for a dataset that contains 1000 instances of 20 variables each, the simulation would, at a maximum, store 1000 $2^n \times 2^n$ density matrices, with n being the number of qubits used. This means that using the full 20 variables, one per qubit, would imply $2^{20} \times 2^{20} \approx 1.1 \times 10^{12}$ complex entries, and at 16 bytes each, that is about 16 TiB per matrix. It was decided that a correlation study over the target variable would be an adequate way to reduce this load. The reasons for this decision are: First, the investigation did not aim to produce a classifier and thus could use a smaller number of variables in the experiments. Second, the variables selected from a correlation study would be the most likely candidates to be chosen as significant by a classifier.

As mentioned previously, however, most of the variables in this dataset are not numeric and thus a popular correlation analysis such as the Pearson correlation is not an adequate choice to evaluate variable correlation. For starters, for a nominal variable the Pearson coefficient relies on an arbitrary choice for the label and since the proportions of the target variable in the dataset for each category are not the same across the board, a change in category label implies a change in the coefficient. Moreover, the Pearson coefficient identifies linear relationships between variables well but fails for nonlinear cases.

For these reasons it was decided to use the PhiK method [10] to select the variables which show a consistently stronger correlation with the target variable. PhiK is a stronger candidate as it works well for both numerical and categorical variables and it is able to identify nonlinear relationships between them.

We used Baak et al.'s PhiK library to measure the correlation and significance between each variable and the target and recorded the results in Table 1. It is important to note that the library returns, for a pair of variables, two results: the first is ϕ_K which is the measurement of how strong the relationship between the two variables is. This result is calculated by calibrating a χ^2 test statistic to a contingency table (populated with the categorical variables or binned numerical variables) with respect to a bivariate normal distribution. This calibration returns a ϕ_K with range of 0 (no association) to 1 (complete association). Since ϕ_K depends on the counts of the contingency table, renumbering the categories no longer affects the end result. The second result is Z which is a conversion of the p-value from the independence test into a number of standard deviations. A higher Z provides more evidence that the variable is not independent from the target.

For the variable selection we chose to use Z because, although ϕ_K measures the correlation between variables, the significance Z of that correlation indicates whether that correlation is significant enough to be told apart from chance. The Bonferroni method states that, to ensure that the probability of at least one false positive across k tests (in our

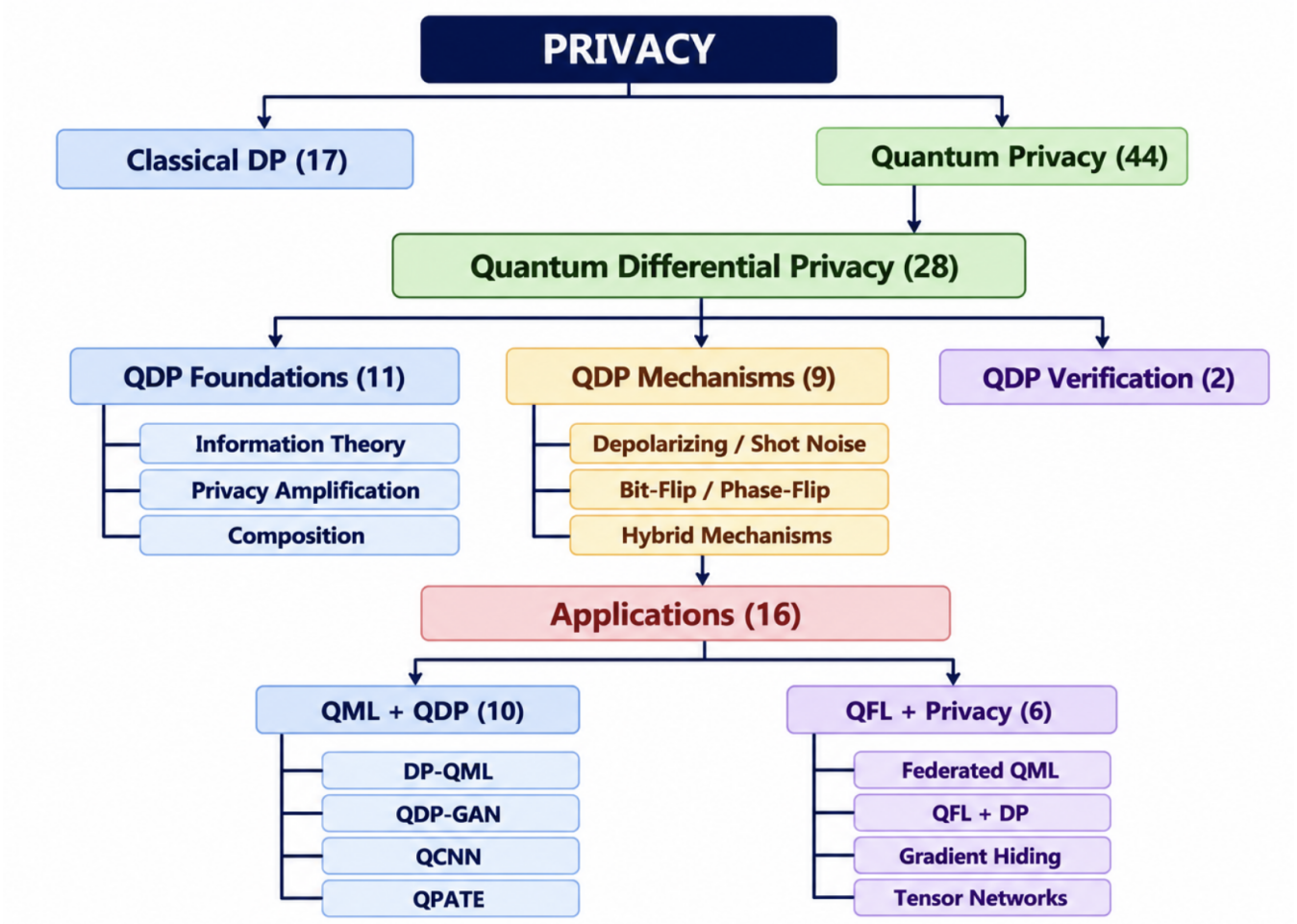


FIGURE 1. State of the art of quantum differential privacy (own elaboration, based on the literature review described above).

TABLE 1. ϕ_K correlation of each feature with the target and its statistical significance with a cutoff of $Z > 2.81$.

Feature	ϕ_K	Z	Significant
status_account	0.512	10.965	Yes
credit_history	0.197	6.912	Yes
month_duration	0.285	5.648	Yes
status_savings	0.147	5.283	Yes
credit_amount	0.241	4.388	Yes
collateral	0.217	4.009	Yes
purpose	0.204	3.715	Yes
housing	0.077	3.601	Yes
years_employment	0.098	3.053	Yes
other_installment_plans	0.063	2.891	Yes
is_foreign_worker	0.109	2.545	No
age	0.142	2.177	No
status_and_sex	0.123	1.953	No
secondary_obligor	0.041	1.835	No
payment_to_income_ratio	0.075	1.045	No
telephone	0.021	0.689	No
n_credits	0.000	0.043	No
job	0.000	−0.390	No
residence_since	0.000	−1.070	No
n_guarantors	0.000	$-\infty$	No

case $k = 20$ for 20 variables against the target) is at most $\alpha = 0.05$, each individual test must use a significance level of $\alpha' = \alpha/k = 0.05/20 = 0.0025$ [11]. Here a false positive is a variable that is independent of the target but is selected anyway. Since $\alpha' = 0.0025$ corresponds to $Z \approx 2.81$, we used $Z > 2.81$ as the cutoff for the selection of the variables.

As the reader can observe in Table 1, the selection took 10 out of the 20 variables, which, as mentioned before, now leaves a maximum of 1000 $2^{10} \times 2^{10}$ matrices as opposed to 1000 $2^{20} \times 2^{20}$ should we use one qubit per variable. We will show how we lowered that number in Section IV-C. The definition and type of each variable is recorded in Appendix A.

C. ENCODING OF CLASSICAL VARIABLES INTO QUANTUM

Having chosen a reduced set of variables from the dataset (10 from the original 20), the next step was to encode the classical information from each variable into qubits which could be used in the quantum simulation. For this it is important to note that the variables were divided into two groups, each with its own encoding:

1) Categorical Variables

These are categorical variables whose categories do not follow a natural ranking. An example of this is the variable `purpose`, which has ten categories (car (new), vacation, education, among others). As the reader can observe, these categories do not follow a natural order, as there is no way to tell if, say, car goes before or after education. Due to this, it was decided to separate them from the categorical (ordinal) variables that will be mentioned later on.

The proposed encoding for these types of variables is given by Definition 4.1.

Definition 4.1: (Method for encoding categorical variables). For a categorical variable with K categories, when encoding the label $c \in \{0, \ldots, K-1\}$, the resulting encoded qubit state $|\psi\rangle$ is given by the equation

$$|\psi\rangle = RZ(\phi)H\,|0\rangle \quad \text{with} \quad \phi = \frac{2\pi c}{K}$$

This encoding ends up with a qubit state in one of K equally distributed points across the equator of the Bloch sphere. We choose to place the qubit on the equator because, from Definition 2.6, we can express τ in terms of the angle between two points of a circle Δ and the radius of that circle as $r \sin \frac{\Delta}{2}$. With this expression the reader can note that the minimum distance τ is maximized for all other angles on the sphere when the radius is maximum, which happens at the equator of the sphere. Moreover, one could argue that having a smaller θ (i.e., the angle of rotation about the y-axis of the Bloch sphere) would reduce ϵ, and one would be right if not for the fact that it comes with the tradeoff of making the categories less distinguishable. In the case where $\theta = 0$, any rotation over the z-axis would be invalidated, since without a rotation over the y-axis it would only add a global phase.

This method assures then that τ between two categories is at least $\sin\left(\frac{\pi}{K}\right)$. What this means in theory is that the privacy in the local setting will not change the final ϵ, as regardless of the labels given to any category, the angle between the labels themselves might change, but the possible angles between different categories will remain the same. However, what this encoding does not prevent is the fact that different categories can have different proportions. The labels given to each category can impact the distance τ between the dataset and that same dataset without a person included, which, as will be explained in Section V, is crucial when computing ϵ_{exp} and ϵ_{theo} and therefore, in theory, can lead to different results. In Appendix B we compiled the c values given to each variable's categories.

2) Numerical and ordinal variables

These are variables which are either ranges of an amount (ordinal) or numerical. An example of these variables is `month_duration`, which gives the loan's duration in months. A loan that goes for 4 months is clearly shorter than a loan that goes for 72, thus it is important to preserve that order. The encoding method which we believed would be able to preserve this order was dense angle encoding [12].

The question that remained, however, was: which variables were going to be encoded together? To answer this question we first created all 15 possible arrangements of the 5 numerical and ordinal variables into two pairs and a single variable. Then, we once again used the PhiK method to analyze the correlation between each pair of variables which is documented in Table 5 (Appendix C). From this small investigation it was concluded that the combination of variables that yielded the smallest correlation in the worst case was `status_account` as the lone variable, `status_savings` and `month_duration` as the first pair, and `years_employment` and `credit_amount` as the second pair. We chose this combination as the smallest correlation between the worst pair would imply that the qubits carried as little redundant information within each qubit as possible.

Before the encoding definition, we defined a function **minmax**(min, max, a), which normalizes a value $a \in [min, max]$ to $a \in [0, \pi]$.

Definition 4.2: (Min-max normalization). Given a value $a \in [min, max]$ with $min, max \in \mathbb{R}$ we define the function **minmax** as

$$minmax(min, max, a) = \pi \times \frac{a - min}{max - min}$$

With this definition the method to encode numerical and ordinal variables is dense angle encoding [12], which we define as:

Definition 4.3: (Method for encoding numerical and ordinal variables with dense angle encoding). For an ordinal variable with K categories and a numerical variable with range $[n_{Min}, n_{Max}]$, when encoding the ordinal variable with label $c \in \{0, \ldots, K-1\}$ and the numerical variable with value $n \in [n_{Min}, n_{Max}]$, the resulting dense angle encoding is given by

$$\begin{aligned} |\psi\rangle &= RZ(\phi)RY(\theta)\,|0\rangle\,, \\ \text{with } \phi &= minmax(n_{Min}, n_{Max}, n), \\ \text{and } \theta &= minmax(-0.5, K-0.5, c) \end{aligned}$$

Because of the pairing selection, the definition only needs to cover the case where an ordinal variable is paired with a numerical one. Thus it is important to note that the reason **minmax** is calibrated from -0.5 to $K - 0.5$ for the ordinal variables is so that the resulting categories cannot reach the poles, which would result in the gate RZ being invalidated. This calibration then sets each category into the middle of a bin of length $\frac{\pi}{K}$. An exception to this definition is the lone variable `status_account`, since it does not need this calibration as there is no RZ gate to invalidate, so the range for **minmax** is $[0, K-1]$.

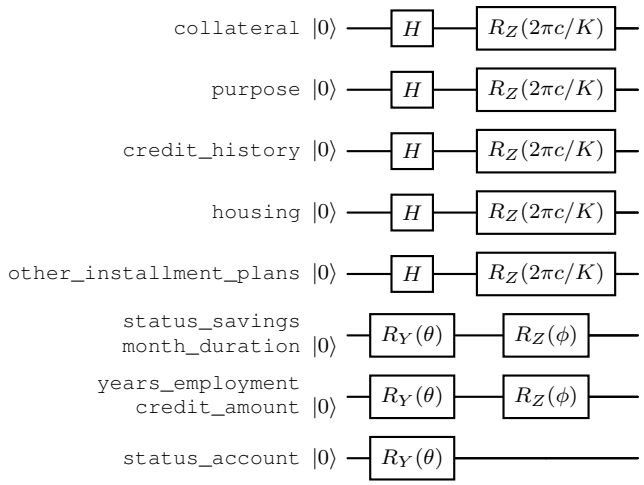


FIGURE 2. 8-qubit encoding circuit for the 10 selected variables

As the reader can observe, thanks to dense angle encoding we can actually reduce the number of qubits necessary for encoding the full 10 variables to 8 which helps reduce the computational toll from the original $2^{10} \times 2^{10}$ matrices with a 1:1 qubit-to-variable encoding to $2^8 \times 2^8$ matrices. The final quantum circuit can be seen in Figure 2.

V. FINDING THE EXPERIMENTAL PRIVACY BUDGET

Before starting with the in-depth explanation of each experiment it is important to mention the noise parameters chosen for all the different mechanisms mentioned in Section II; these are compiled in Table 2. All of these parameters are just that, parameters, so the selection was made mainly as a way to, firstly, execute all the tests with the same parameters and thus allow for a comparison between experiments and, secondly, abide by the restrictions imposed on λ and γ by Theorem 2.2 and on η by Theorem 2.1. Furthermore, due to the nature of the PAD mechanism, it was important to choose a $\lambda \in (0, 0.5)$ to allow the mechanism to behave differently from GAD and DEP.

TABLE 2. Selection of parameters for noise mechanisms

Mechanism	Parameters
Depolarizing	$p = 0.5$
Generalized Amplitude Damping	$\eta = 0.5, \gamma = 0.5$
Phase-Amplitude Damping	$\eta = 0.5, \gamma = 0.5, \lambda = 0.25$

A. CENTRAL SETTING

The first experiment that was carried out was finding the ϵ_{theo} and ϵ_{exp} in a central setting for each qubit k of the dataset (recall the encoding from Section IV-C). In other words, we wanted to see what privacy guarantee each of the mechanisms $\mathcal{M}_{GAD}$, $\mathcal{M}_{PAD}$ and $\mathcal{M}_{DEP}$ provided. In order to accomplish this, we designed the pipeline documented in Figure 3.

The pipeline begins by calculating some values that will be used during the leave-one-out loop: encoding the full column using the methodology provided in Section IV, which will become our set $\{\rho_i^k\}$, containing the states of each person in the dataset for that column. That complete set will be aggregated into a single matrix ρ^k, which will hold an average of all the participants of the dataset for the column k. Then noise from the selected noise mechanism will be applied to that density matrix, resulting in the matrix ρ'^k. With those two matrices calculated, the pipeline will begin a loop that iterates over each participant of the dataset in order to compare the original against a leave-one-out version of the dataset where the row j is removed.

For each row j, the process is as follows: aggregate the set $\{\rho_i^k\}$ with the row j removed, which results in the density matrix σ_j^k, the aggregation of all participants except j. From σ_j^k both $\sigma_j'^k$ and d_j will be calculated. $\sigma_j'^k$ is the density matrix that results from applying the selected noise mechanism to σ_j^k, and d_j is the trace distance between ρ^k and σ_j^k, which has to be calculated with the pre-noise matrices as per Theorems 2.1 through 2.3. Lastly, it will calculate ϵ_{exp}^j, which is the ϵ_{exp} for the j-th row and corresponds to the maximum of the values that Algorithm 1 returns in both directions.

With all those values, the only remaining step of the pipeline is to input them into each method for finding ϵ. On the theoretical side, the pipeline will input the maximum trace distance and the noise mechanism's parameters into one of the theorems defined in Section II (for the depolarizing mechanism $D = 2$, as we are calculating on a per-qubit basis) for the resulting ϵ_{theo} of the column. In the case of ϵ_{exp}, it is the maximum ϵ_{exp}^j over all rows j, as that is the maximum privacy loss that the dataset can offer for that noise mechanism.

With the ϵ_{exp}^k and ϵ_{theo}^k calculated for each column k of the dataset, the only thing that remains is to add them to find the upper bound of ϵ_{exp} and ϵ_{theo} of the dataset. In the pipeline, the curator would release the noisy density matrices for the 8 qubits

$$\bigotimes_{k=0}^{7} \rho'^k = \bigotimes_{k=0}^{7} \mathcal{M}(\rho^k).$$

Because of Lemma 5.1, this is equal to

$$\Big(\bigotimes_{k=0}^{7} \mathcal{M}\Big)\Big(\bigotimes_{k=0}^{7} \rho^k\Big) = \mathcal{M}_{[8]}(\rho^0, \ldots, \rho^7)$$

which is the combination defined in Corollary 2.1, and thus the addition is valid.

Lemma 5.1: For $i = 1, \ldots, k$, let $\mathcal{E}_i(\rho) = \sum_a E_{i,a}\, \rho\, E_{i,a}^\dagger$ be a quantum operation on $\mathcal{H}_i$ and ρ_i a density operator on $\mathcal{H}_i$. Then

$$\Big(\bigotimes_{i=1}^{k} \mathcal{E}_i\Big)\Big(\bigotimes_{i=1}^{k} \rho_i\Big) = \bigotimes_{i=1}^{k} \mathcal{E}_i(\rho_i).$$

For readability, the proof of Lemma 5.1 is deferred to Appendix D.

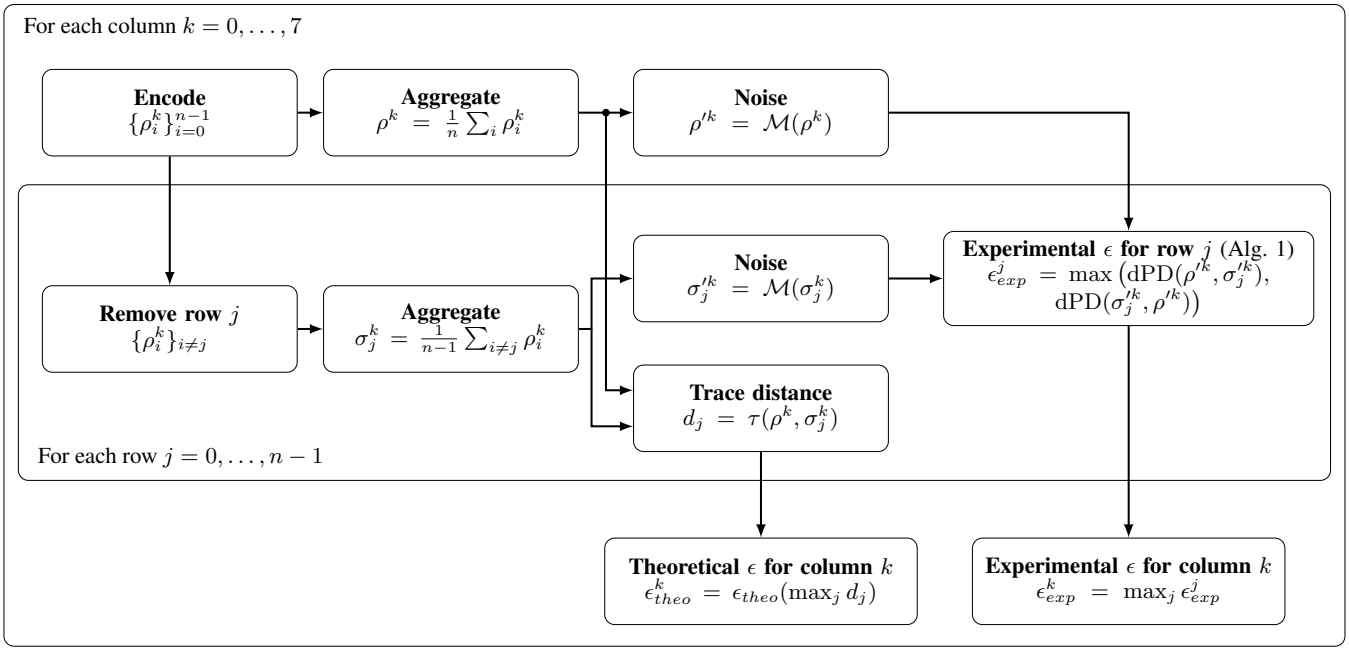


FIGURE 3. Experimental pipeline for finding ϵ_{exp} and ϵ_{theo}

In the case of the depolarizing mechanism, we can actually modify the pipeline by removing the need to encode each column individually and applying the noise to the full 8 columns, since Theorem 2.3 allows it. This, however, is classified as a separate experiment, since both the aggregation and the release that the curator would perform are over the full 8-qubit matrix (not per qubit, as was explained), thus not allowing for a comparison against the original pipeline. Later on, in Section VIII, we will discuss how these methods yield very different privacy results.

B. LOCAL SETTING

We have now considered an experiment where a trusted curator applies the noise to the data and releases the data. This experiment, however, tries to answer the question of what would happen if it were each individual person applying the noise, the local setting.

In the central setting we defined neighbors as two aggregated density matrices, the one for the full 1000 ρ rows and the leave-one-out σ. In this setting (inspired by [5, Def. 12.1] as mentioned in Section II), however, since it is just one person at a time who is privatizing their data, it would not make sense to compare two different people as neighbors, as in reality what the attacker would want to know is "what was person A's original category?". With this idea, this experiment defines the neighbors as the possible states that each label is encoded into, and because of Definition 2.15 we can compare two states as neighbors by their trace distance τ. This experiment, however, limits the variables we can encode. As mentioned in Subsection IV-C the numerical variables need to be normalized within the Bloch sphere with the minimum and maximum values of the variables. In the local setting a single person has no way to access that data and thus cannot encode it in that manner. With that limitation in mind, this setting calculates the ϵ for all the categorical variables and the sole variable `status_account` since the other two ordinal variables are accompanied in the same qubit by a numerical variable and thus cannot be encoded independently. Figure 4 shows the pipeline for the local setting experiment.

The experiment begins by defining two sets $\{\rho_c\}$ which are all the density matrices that result from encoding the c-th label and $\{\rho'_c\}$ which is the set of all the density matrices ρ_c that have been passed over the noise mechanism $\mathcal{M}$. Then, for each pair of labels $\{c, c'\}, c < c'$, we calculate both the trace distance between the original noiseless density matrices ρ_c and $\rho_{c'}$ and the ϵ_{exp} which is the same procedure as the central setting just with inputs ρ'_c and $\rho'_{c'}$. Then, at the end, we make the same search for the maximum ϵ_{theo} and ϵ_{exp} of all the analyzed pairs and that is the final result of the variable.

VI. MEASUREMENT AND CLASSICAL RELEASE OF CATEGORICAL VARIABLES

La explicación de la medicion de las variables categoricas como forma de mostrar el ruido en acción. Es el resto de la presentación de semana 10 puesto en texto

VII. EXPERIMENTAL RESULTS

A. FINDING THE EXPERIMENTAL PRIVACY BUDGET

1) Central Setting

Resultados del experimento central

2) Local Setting

Resultados del experimento local

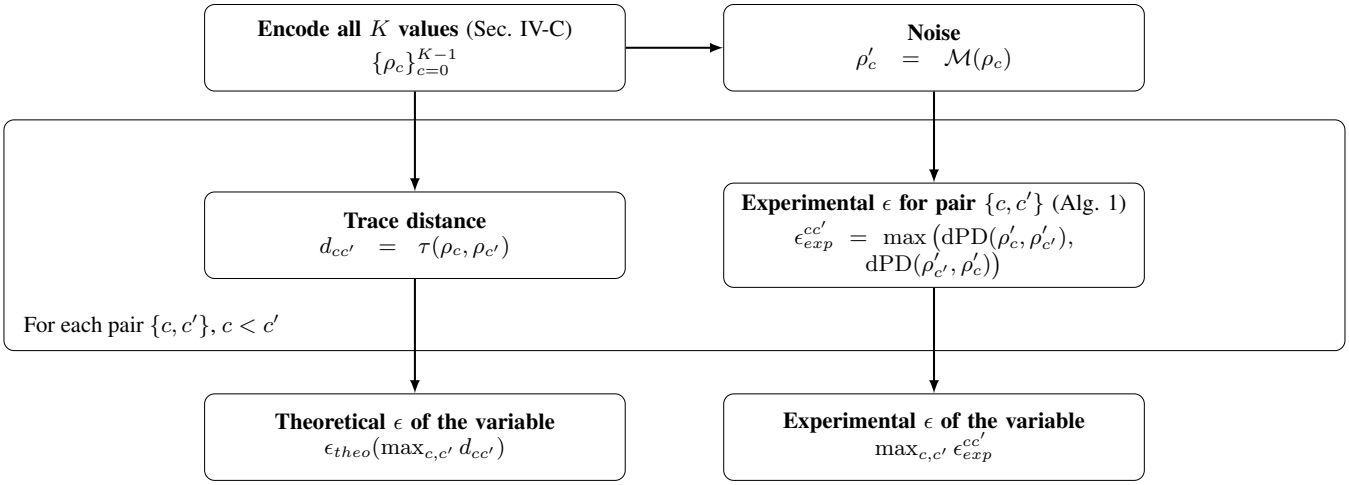


FIGURE 4. Pipeline for the local setting experiment for finding ϵ for a categorical or ordinal variable

B. MEASUREMENT AND CLASSICAL RELEASE OF CATEGORICAL VARIABLES

Resultados de las mediciones con las distribuciones de probabilidad

VIII. DISCUSSION AND CONCLUSION

En esta seccion voy a hablar de limitaciones encontradas dentro de los experimentos.

Entre ellas:

La dualidad entre C y L para la privacidad. (pros y contras) Que limitaciones tiene el experimento (analisis conjunto)

Tambien el analisis de los resultados de los mismos y que implicaciones tienen estos frente a la privacidad diferencial tanto clasica como cuantica.

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APPENDIX A DESCRIPTION OF SELECTED FEATURES

Table 3 compiles the name, type and description of the variables that were selected for use in the quantum experiments.

APPENDIX B DESCRIPTION OF LABEL ASSIGNED TO EACH CATEGORICAL VARIABLE

Table 4 compiles the labels given to each variable's categories using the scikit-learn Python library.

APPENDIX C ϕ_K CORRELATION FOR ALL COMBINATIONS OF NUMERICAL AND ORDINAL VARIABLES

Table 5 shows the ϕ_K correlation results for all combinations of numerical and ordinal variables, as well as their worst correlation between the two pairs.

APPENDIX D PER-QUBIT NOISE AS A COMBINED MECHANISM

In this appendix we prove Lemma 5.1 which is used in Section V in order to allow for the addition of the ϵ^k_{theo} and ϵ^k_{exp} for the k qubits by demonstrating that the tensor product of the outputs, $\bigotimes_i \mathcal{E}_i(\rho_i)$, is the same as the tensor product of the operations applied to the tensor product of the states, $(\bigotimes_i \mathcal{E}_i)(\bigotimes_i \rho_i)$.

Lemma D.1: Let X_i and U_i be linear operators on a vector space $\mathcal{H}_i$, for $i = 1, \ldots, k$. Then

$$\Big(\bigotimes_{i=1}^{k} X_i\Big)\Big(\bigotimes_{i=1}^{k} U_i\Big) = \bigotimes_{i=1}^{k} X_i U_i.$$

Proof: Two linear operators are equal if and only if they act in the same way on every vector of the space. The product vectors $e_1 \otimes \cdots \otimes e_k$, with each e_i taken from a basis of $\mathcal{H}_i$, span $\bigotimes_i \mathcal{H}_i$, and both sides are linear, so it is enough to compare them on these vectors. By the definition of the tensor product of operators, $(\bigotimes_i T_i)(\bigotimes_i v_i) = \bigotimes_i T_i v_i$ [13, Eq. (1.8)]. Therefore

$$\begin{aligned}\Big(\bigotimes_i X_i\Big)\Big(\bigotimes_i U_i\Big)\Big(\bigotimes_i e_i\Big) &= \Big(\bigotimes_i X_i\Big)\Big(\bigotimes_i U_i e_i\Big)\\ &= \bigotimes_i X_i U_i e_i\\ &= \Big(\bigotimes_i X_i U_i\Big)\Big(\bigotimes_i e_i\Big),\end{aligned}$$

so both operators agree on every product vector, and hence on all of $\bigotimes_i \mathcal{H}_i$. ■

Proposition D.1: Let A_i be a linear operator and ρ_i a density operator on $\mathcal{H}_i$, for $i = 1, \ldots, k$. Then

$$\Big(\bigotimes_{i=1}^{k} A_i\Big)\Big(\bigotimes_{i=1}^{k} \rho_i\Big)\Big(\bigotimes_{i=1}^{k} A_i\Big)^{\dagger} = \bigotimes_{i=1}^{k} A_i \rho_i A_i^{\dagger}.$$

Proof: The adjoint of a tensor product is the tensor product of the adjoints, $(\bigotimes_i A_i)^{\dagger} = \bigotimes_i A_i^{\dagger}$ [3]. Applying Lemma D.1 twice,

$$\begin{aligned}\Big(\bigotimes_i A_i\Big)\Big(\bigotimes_i \rho_i\Big)\Big(\bigotimes_i A_i^{\dagger}\Big) &= \Big(\bigotimes_i A_i\Big)\Big(\bigotimes_i \rho_i A_i^{\dagger}\Big)\\ &= \bigotimes_i A_i \rho_i A_i^{\dagger}.\end{aligned}$$

■

Proof of Lemma 5.1: The operation $\bigotimes_i \mathcal{E}_i$ has Kraus operators $E_{1,a_1} \otimes \cdots \otimes E_{k,a_k}$, one for each choice of indices $a_1, \ldots, a_k$. Applying Proposition D.1 to each term, and then the distributivity (bilinearity) of the tensor product over sums,

$$\begin{aligned}&\Big(\bigotimes_i \mathcal{E}_i\Big)\Big(\bigotimes_i \rho_i\Big)\\ &= \sum_{a_1,\ldots,a_k} \Big(\bigotimes_i E_{i,a_i}\Big)\Big(\bigotimes_i \rho_i\Big)\Big(\bigotimes_i E_{i,a_i}\Big)^{\dagger}\\ &= \sum_{a_1,\ldots,a_k} \bigotimes_i E_{i,a_i}\, \rho_i\, E_{i,a_i}^{\dagger}\\ &= \bigotimes_i \sum_{a_i} E_{i,a_i}\, \rho_i\, E_{i,a_i}^{\dagger} = \bigotimes_i \mathcal{E}_i(\rho_i).\end{aligned}$$

■

TABLE 3. Description of the selected variables.

Name	Type	Description
status_account	Categorical (Ordinal)	Status of the existing checking account, in DM (Deutsche Marks)
month_duration	Numerical	Duration of the credit in months
credit_amount	Numerical	Credit amount
collateral	Categorical	Possible collateral for the loan
purpose	Categorical	Purpose of the credit
credit_history	Categorical	Credit history of the applicant
housing	Categorical	Current housing situation (rented, owned, or free)
years_employment	Categorical (Ordinal)	Length of time in current employment
status_savings	Categorical (Ordinal)	Status of the savings account, in DM (Deutsche Marks)
other_installment_plans	Categorical	Whether the client has other installment plans

TABLE 4. Assignment of the label given to each categorical variable's category

Variable	Category	Label
collateral	car	0
	none	1
	real estate	2
	savings agreement/life insurance	3
purpose	business	0
	car (new)	1
	car (used)	2
	domestic appliances	3
	education	4
	furniture/equipment	5
	others	6
	radio/television	7
	repairs	8
	retraining	9
credit_history	all credits at this bank paid back duly	0
	critical account/ other credits existing (not at this bank)	1
	delay in paying off in the past	2
	existing credits paid back duly till now	3
	no credits taken/ all credits paid back duly	4
housing	for free	0
	own	1
	rent	2
other_installment_plans	bank	0
	none	1
	stores	2

TABLE 5. Comparison of variable pairs and PhiK metrics

			ϕ_K		
Alone	First pair	Second pair	First pair	Second pair	Worst
status_account	status_savings, month_duration	years_employment, credit_amount	0.000	0.043	0.043
credit_amount	status_account, years_employment	status_savings, month_duration	0.088	0.000	0.088
status_savings	status_account, month_duration	years_employment, credit_amount	0.091	0.043	0.091
status_account	status_savings, credit_amount	years_employment, month_duration	0.092	0.065	0.092
years_employment	status_account, month_duration	status_savings, credit_amount	0.091	0.092	0.092
month_duration	status_account, years_employment	status_savings, credit_amount	0.088	0.092	0.092
status_savings	status_account, credit_amount	years_employment, month_duration	0.114	0.065	0.114
years_employment	status_account, credit_amount	status_savings, month_duration	0.114	0.000	0.114
credit_amount	status_account, month_duration	status_savings, years_employment	0.091	0.150	0.150
month_duration	status_account, credit_amount	status_savings, years_employment	0.114	0.150	0.150
credit_amount	status_account, status_savings	years_employment, month_duration	0.200	0.065	0.200
month_duration	status_account, status_savings	years_employment, credit_amount	0.200	0.043	0.200
status_account	status_savings, years_employment	credit_amount, month_duration	0.150	0.689	0.689
status_savings	status_account, years_employment	credit_amount, month_duration	0.088	0.689	0.689
years_employment	status_account, status_savings	credit_amount, month_duration	0.200	0.689	0.689

ACKNOWLEDGMENTS

Esta seccion solo para la tesis o no se si pueda ir en el paper tambien

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